Lab 5 Report:

APIs & Web Services I

Step 0 - Setup

0a Install Dependencies

Prepare the Python virtual environment by installing Flask and other required data processing libraries.

Since I'm using an Anaconda virtual environment, I don't need to create a new one; I can just install Flask and the other necessary data processing libraries directly.

```
pip install flask kafka-python-ng pyspark==3.5.3
```

```
(base) PS C:\Users\xback20040219> conda activate bigdata
(bigdata) PS C:\Users\xback20040219> cd "D:\EFREI\Data_Engineering\LAB\Lab5"
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> pip install flask kafka-python-ng pyspark==3.5.3
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: kafka-python-ng in d:\app\anaconda\envs\bigdata\lib\site-packages (2.2.3)
Requirement already satisfied: pyspark==3.5.3 in d:\app\anaconda\envs\bigdata\lib\site-packages (3.5.3)
Requirement already satisfied: py4j==0.10.9.7 in d:\app\anaconda\envs\bigdata\lib\site-packages (from pyspark==3.5.3) (0.10.9.7)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Requirement already satisfied: importlib-metadata>=3.6.0 in d:\app\anaconda\envs\bigdata\lib\site-packages (from flask) (8.7.1)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: jinja2>=3.1.2 in d:\app\anaconda\envs\bigdata\lib\site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in d:\app\anaconda\envs\bigdata\lib\site-packages (from flask) (3.0.3)
Collecting werkzeug>=3.1.0 (from flask)
  Downloading werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: colorama in d:\app\anaconda\envs\bigdata\lib\site-packages (from click>=8.1.3->flask) (0.4.6)
Requirement already satisfied: zipp>=3.20 in d:\app\anaconda\envs\bigdata\lib\site-packages (from importlib-metadata>=3.6.0->flask) (3.23.0)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.1.8-py3-none-any.whl (98 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Installing collected packages: werkzeug, itsdangerous, click, blinker, flask
Successfully installed blinker-1.9.0 click-8.1.8 flask-3.1.3 itsdangerous-2.2.0 werkzeug-3.1.8
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5>
```

0b Verify Data Lake Has Data

Ensure that the curated Parquet files generated from the previous lab4 exist, and apply necessary Python 3.12+ compatibility fixes.

```
Get-ChildItem -Path "D:\EFREI\Data_Engineering\LAB\Lab4\datalake\curated\domain=iot" -
ErrorAction SilentlyContinue
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> Get-ChildItem -Path "D:\EFREI\Data_Engineering\LAB\Lab4\datalake\curated\domain=iot" -ErrorAction SilentlyContinue


    目录: D:\EFREI\Data_Engineering\LAB\Lab4\datalake\curated\domain=iot


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2026/5/14     11:55                sensor_type=humidity
d-----         2026/5/14     11:55                sensor_type=pressure
d-----         2026/5/14     11:55                sensor_type=temperature
d-----         2026/5/14     11:55                _spark_metadata
```

0c Project Structure

Create the directory structure to separate the API routing logic from the utility functions.

```
New-Item -ItemType Directory -Path "sensor_api" -Force
# We will create these files inside:
# sensor_api/app.py (Flask application)
# sensor_api/kafka_utils.py (Kafka consumer helper)
# sensor_api/lake_utils.py (Parquet query helper)
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> New-Item -ItemType Directory -Path "sensor_api" -Force


    目录: D:\EFREI\Data_Engineering\LAB\Lab5


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2026/5/15      9:24                sensor_api
```

Create the kafka_utils.py

```
# sensor_api/kafka_utils.py
from kafka import KafkaProducer, KafkaConsumer
import json

KAFKA_BROKER = 'localhost:9092'
TOPIC_NAME = 'sensor-events'

def publish_reading(reading):
    """Publish a new reading to Kafka and return metadata."""
    producer = KafkaProducer(
        bootstrap_servers=[KAFKA_BROKER],
        value_serializer=lambda v: json.dumps(v).encode('utf-8')
    )
    future = producer.send(TOPIC_NAME, reading)
    result = future.get(timeout=10)
    return {"partition": result.partition, "offset": result.offset}

def get_latest_readings(sensor_type, n=1):
    """Consume recent messages and return the latest 'n' readings for a sensor type."""
    consumer = KafkaConsumer(
        TOPIC_NAME,
        bootstrap_servers=[KAFKA_BROKER],
        auto_offset_reset='earliest',
        enable_auto_commit=False,
        value_deserializer=lambda x: json.loads(x.decode('utf-8', errors='ignore')),
        consumer_timeout_ms=5000
    )

    records = []
    for message in consumer:
        val = message.value
        if isinstance(val, dict) and val.get("sensor") == sensor_type:
```

```
            records.append(val)

    records.sort(key=lambda r: r.get("timestamp") or 0, reverse=True)

    return records[:n]
```

Lab5 > sensor_api > kafka_utils.py > ...

```python
# sensor_api/kafka_utils.py
from kafka import KafkaProducer, KafkaConsumer
import json

KAFKA_BROKER = 'localhost:9092'
TOPIC_NAME = 'sensor-events'

def publish_reading(reading):
    """Publish a new reading to Kafka and return metadata."""
    producer = KafkaProducer(
        bootstrap_servers=[KAFKA_BROKER],
        value_serializer=lambda v: json.dumps(v).encode('utf-8')
    )
    future = producer.send(TOPIC_NAME, reading)
    result = future.get(timeout=10)
    return {"partition": result.partition, "offset": result.offset}

def get_latest_readings(sensor_type, n=1):
    """Consume recent messages and return the latest 'n' readings for a sensor type."""
    consumer = KafkaConsumer(
        TOPIC_NAME,
        bootstrap_servers=[KAFKA_BROKER],
        auto_offset_reset='earliest',
        enable_auto_commit=False,
        value_deserializer=lambda x: json.loads(x.decode('utf-8', errors='ignore')),
        consumer_timeout_ms=5000
    )

    records = []
    for message in consumer:
        val = message.value
        if isinstance(val, dict) and val.get("sensor") == sensor_type:
            records.append(val)

    records.sort(key=lambda r: r.get("timestamp") or 0, reverse=True)

    return records[:n]
```

Create the lake_utils.py

```python
# sensor_api/lake_utils.py
import os
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
os.environ['HADOOP_HOME'] = r"D:\EFREI\Data_Engineering\LAB\Lab3\hadoop"
DATALAKE_PATH = r"D:\EFREI\Data_Engineering\LAB\Lab4\datalake\curated\domain=iot"
def _get_spark():
    """Initialize SparkSession."""
    try:
        spark = SparkSession.builder \
            .appName("SensorAPI") \
            .master("local[*]") \
            .getOrCreate()
```

```
        return spark
    except Exception as e:
        print(f"Failed to start Spark: {e}")
        return None
def get_sensor_types():
    """Read Parquet files to get all distinct sensor types."""
    try:
        spark = _get_spark()
        if not spark:
            return []
        df = spark.read.parquet(DATALAKE_PATH)
        types = [row[0] for row in df.select("sensor_type").distinct().collect()]
        return types
    except Exception as e:
        print(f"Error reading datalake for sensor types: {e}")
        return []

def get_statistics(sensor_type, days=7):
    """Mock statistics retrieval from Parquet."""
    try:
        spark = _get_spark()
        if not spark:
            return []

        df = spark.read.parquet(DATALAKE_PATH)
        stats_df = df.filter(col("sensor_type") == sensor_type).limit(days)

        return [row.asDict() for row in stats_df.collect()]
    except Exception as e:
        print(f"Error reading stats: {e}")
        return []
```

Lab5 > sensor_api > lake_utils.py > _get_spark

```python
# sensor_api/lake_utils.py
import os
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
os.environ['HADOOP_HOME'] = r"D:\EFREI\Data_Engineering\LAB\Lab3\hadoop"
DATALAKE_PATH = r"D:\EFREI\Data_Engineering\LAB\Lab4\datalake\curated\domain=iot"
def _get_spark():
    """Initialize SparkSession."""
    try:
        spark = SparkSession.builder \
            .appName("SensorAPI") \
            .master("local[*]") \
            .getOrCreate()
        return spark
    except Exception as e:
        print(f"Failed to start Spark: {e}")
        return None
def get_sensor_types():
    """Read Parquet files to get all distinct sensor types."""
    try:
        spark = _get_spark()
        if not spark:
            return []
        df = spark.read.parquet(DATALAKE_PATH)
        types = [row[0] for row in df.select("sensor_type").distinct().collect()]
        return types
    except Exception as e:
        print(f"Error reading datalake for sensor types: {e}")
        return []

def get_statistics(sensor_type, days=7):
    """Mock statistics retrieval from Parquet."""
    try:
        spark = _get_spark()
        if not spark:
            return []

        df = spark.read.parquet(DATALAKE_PATH)
        stats_df = df.filter(col("sensor_type") == sensor_type).limit(days)

        return [row.asDict() for row in stats_df.collect()]
    except Exception as e:
        print(f"Error reading stats: {e}")
        return []
```

Step 1 - Application Factory and Health Check

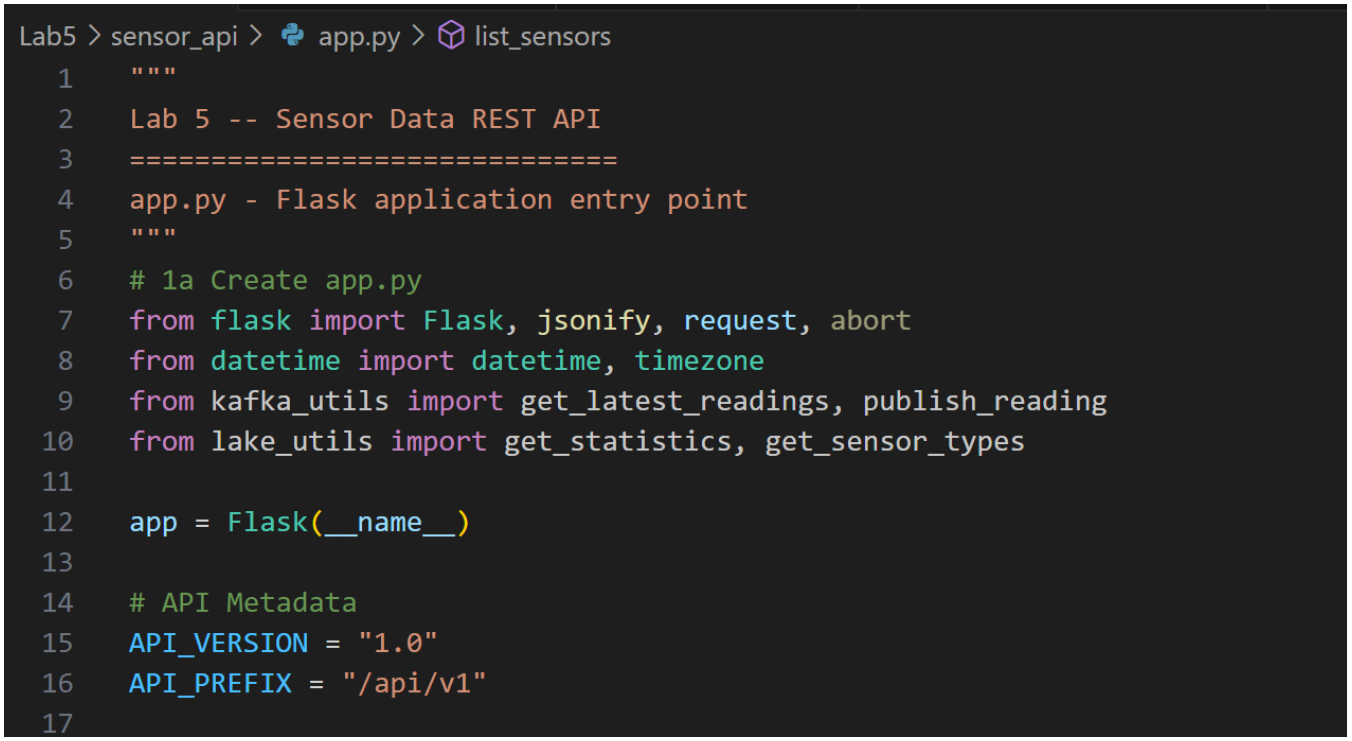
1a Create app.py

Initialize the Flask application entry point, define global metadata, and import necessary utilities.

```python
# sensor_api/app.py
from flask import Flask, jsonify, request, abort
from datetime import datetime, timezone
from kafka_utils import get_latest_readings, publish_reading
from lake_utils import get_statistics, get_sensor_types

app = Flask(__name__)

# API Metadata
API_VERSION = "1.0"
API_PREFIX = "/api/v1"
```



1b Health Check Endpoint

Create a mandatory endpoint used by load balancers and monitoring systems to verify the API is running.

```python
@app.route(f"{API_PREFIX}/health")
def health():
    return jsonify({
        "status": "ok",
        "version": API_VERSION,
        "timestamp": datetime.now(timezone.utc).isoformat(),
        "service": "sensor-data-api",
    }), 200
```

```python
# 1b Health Check Endpoint
@app.route(f"{API_PREFIX}/health")
def health():
    return jsonify({
        "status": "ok",
        "version": API_VERSION,
        "timestamp": datetime.now(timezone.utc).isoformat(),
        "service": "sensor-data-api",
    }), 200
```

Step 2 - Sensor List and Latest Reading Endpoints

2a List All Sensor Types

Create a `GET` endpoint to retrieve the list of all known sensor types from the Parquet curated zone.

```python
@app.route(f"{API_PREFIX}/sensors")
def list_sensors():
    try:
        sensor_types = get_sensor_types()
        return jsonify({
            "status": "success",
            "count": len(sensor_types),
            "data": sensor_types,
        }), 200
    except Exception as exc:
        app.logger.error("list_sensors error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve sensor list.",
        }), 500
```

```python
# 2a List All Sensor Types
@app.route(f"{API_PREFIX}/sensors")
def list_sensors():
    """
    GET /api/v1/ sensors
    Returns the list of all known sensor types .
    Data is read from the Parquet curated zone .
    """
    try:
        sensor_types = get_sensor_types()
        return jsonify({
            "status": "success",
            "count": len(sensor_types),
            "data": sensor_types,
        }), 200
    except Exception as exc:
        app.logger.error("list_sensors error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve sensor list.",
        }), 500
```

2b Get Latest Reading for a Sensor Type

Create an endpoint that extracts a path parameter to query the most recent reading from Kafka.

```python
VALID_SENSORS = {"temperature", "humidity", "pressure"}

@app.route(f"{API_PREFIX}/sensors/<sensor_type>/latest")
def latest_reading(sensor_type: str):
    if sensor_type not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": (f"Unknown sensor type '{sensor_type}'. "
                        f"Valid types: {sorted(VALID_SENSORS)}")
        }), 404
    try:
        reading = get_latest_readings(sensor_type, n=1)
        if not reading:
            return jsonify({
                "status": "error",
                "message": f"No readings available for sensor '{sensor_type}'."
            }), 404

        return jsonify({
            "status": "success",
```

```
            "data": reading[0],
        }), 200
    except Exception as exc:
        app.logger.error("latest_reading error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve latest reading.",
        }), 500
```

```python
# 2b Get Latest Reading for a Sensor Type
VALID_SENSORS = {"temperature", "humidity", "pressure"}

@app.route(f"{API_PREFIX}/sensors/<sensor_type>/latest")
def latest_reading(sensor_type: str):
    """
    GET /api/v1/ sensors /{ sensor_type }/ latest
    Returns the most recent reading from Kafka for the given sensor type .
    Path parameters :
    sensor_type ( str): one of temperature , humidity , pressure
    Returns :
    200 + reading if found
    404 if sensor_type is unknown
    """
    if sensor_type not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": (f"Unknown sensor type '{sensor_type}'. "
                        f"Valid types: {sorted(VALID_SENSORS)}")
        }), 404
    try:
        reading = get_latest_readings(sensor_type, n=1)
        if not reading:
            return jsonify({
                "status": "error",
                "message": f"No readings available for sensor '{sensor_type}'."
            }), 404

        return jsonify({
            "status": "success",
            "data": reading[0],
        }), 200
    except Exception as exc:
        app.logger.error("latest_reading error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve latest reading.",
        }), 500
```

Step 3 - Statistics from the Parquet Data Lake

Create an endpoint to return daily statistics for a sensor type, utilizing query parameters for filtering.

```
@app.route(f"{API_PREFIX}/sensors/<sensor_type>/stats")
```

```
def sensor_stats(sensor_type: str):
    if sensor_type not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": f"Unknown sensor type '{sensor_type}'.",
        }), 404
    try:
        days = int(request.args.get("days", 7))
        if days < 1 or days > 90:
            raise ValueError("days must be between 1 and 90")
    except (ValueError, TypeError) as exc:
        return jsonify({
            "status": "error",
            "message": f"Invalid 'days' parameter: {exc}",
        }), 400

    try:
        stats = get_statistics(sensor_type, days=days)
        return jsonify({
            "status": "success",
            "sensor_type": sensor_type,
            "days": days,
            "count": len(stats),
            "data": stats,
        }), 200
    except Exception as exc:
        app.logger.error("sensor_stats error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve statistics.",
        }), 500
```

```python
# 3 Statistics from the Parquet Data Lake
@app.route(f"{API_PREFIX}/sensors/<sensor_type>/stats")
def sensor_stats(sensor_type: str):
    """
    GET /api/v1/ sensors /{ sensor_type }/ stats
    Returns daily statistics for a sensor type .
    Query parameters ( optional ):
    days (int ): number of recent days ( default : 7)
    Example :
    GET /api/v1/ sensors / temperature / stats ? days =3
    """
    if sensor_type not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": f"Unknown sensor type '{sensor_type}'.",
        }), 404
    try:
        days = int(request.args.get("days", 7))
        if days < 1 or days > 90:
            raise ValueError("days must be between 1 and 90")
    except (ValueError, TypeError) as exc:
        return jsonify({
            "status": "error",
            "message": f"Invalid 'days' parameter: {exc}",
        }), 400

    try:
        stats = get_statistics(sensor_type, days=days)
        return jsonify({
            "status": "success",
            "sensor_type": sensor_type,
            "days": days,
            "count": len(stats),
            "data": stats,
        }), 200
    except Exception as exc:
        app.logger.error("sensor_stats error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to retrieve statistics.",
        }), 500
```

Step 4 - POST: Write a Reading to Kafka

Create a POST endpoint to publish new sensor readings to the Kafka topic, handling JSON request bodies.

```
import json
REQUIRED_FIELDS = {"sensor", "value"}

@app.route(f"{API_PREFIX}/readings", methods=["POST"])
def create_reading():
    body = request.get_json(silent=True)
    if body is None:
        return jsonify({
            "status": "error",
            "message": "Request body must be valid JSON. Set Content-Type:
application/json."
        }), 400

    missing = REQUIRED_FIELDS - set(body.keys())
    if missing:
        return jsonify({
            "status": "error",
            "message": f"Missing required fields: {missing}"
        }), 400

    sensor = body["sensor"]
    if sensor not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": f"Invalid sensor type '{sensor}'."
        }), 422

    try:
        value = float(body["value"])
    except (ValueError, TypeError):
        return jsonify({
            "status": "error",
            "message": "'value' must be a number."
        }), 422

    reading = {
        "sensor": sensor,
        "value": value,
        "unit": body.get("unit", ""),
        "timestamp": int(datetime.now(timezone.utc).timestamp() * 1000),
        "source": "api-v1",
        "anomaly": False,
    }

    try:
        metadata = publish_reading(reading)
        return jsonify({
            "status": "success",
            "message": "Reading published to Kafka.",
```

```
            "data": {
                "reading": reading,
                "partition": metadata["partition"],
                "offset": metadata["offset"],
            }
        }), 201
    except Exception as exc:
        app.logger.error("create_reading error: %s", exc)
        return jsonify({
            "status": "error",
            "message": "Failed to publish reading."
        }), 500
```

```
# 4 POST: Write a Reading to Kafka
import json
REQUIRED_FIELDS = {"sensor", "value"}

@app.route(f"{API_PREFIX}/readings", methods=["POST"])
def create_reading():
    """
    POST /api/v1/readings
    Publish a new sensor reading to the Kafka topic.
    Request body ( JSON ):
    {
        "sensor": "temperature",
        "value": 28.5,
        "unit": "C" ( optional )
    }
    Returns :
    201 + published message metadata on success
    400 if body is malformed or missing required fields
    """
    body = request.get_json(silent=True)
    if body is None:
        return jsonify({
            "status": "error",
            "message": "Request body must be valid JSON. Set Content-Type: application/json."
        }), 400

    missing = REQUIRED_FIELDS - set(body.keys())
    if missing:
        return jsonify({
            "status": "error",
            "message": f"Missing required fields: {missing}"
        }), 400

    sensor = body["sensor"]
    if sensor not in VALID_SENSORS:
        return jsonify({
            "status": "error",
            "message": f"Invalid sensor type '{sensor}'."
        }), 422
```

```
170
171     try:
172         value = float(body["value"])
173     except (ValueError, TypeError):
174         return jsonify({
175             "status": "error",
176             "message": "'value' must be a number."
177         }), 422
178
179     reading = {
180         "sensor": sensor,
181         "value": value,
182         "unit": body.get("unit", ""),
183         "timestamp": int(datetime.now(timezone.utc).timestamp() * 1000),
184         "source": "api-v1",
185         "anomaly": False,
186     }
187
188     try:
189         metadata = publish_reading(reading)
190         return jsonify({
191             "status": "success",
192             "message": "Reading published to Kafka.",
193             "data": {
194                 "reading": reading,
195                 "partition": metadata["partition"],
196                 "offset": metadata["offset"],
197             }
198         }), 201
199     except Exception as exc:
200         app.logger.error("create_reading error: %s", exc)
201         return jsonify({
202             "status": "error",
203             "message": "Failed to publish reading."
204         }), 500
```

Step 5 - Error Handlers

Register global handlers to capture unhandled Flask exceptions and enforce JSON-formatted responses.

```
@app.errorhandler(404)
def not_found(error):
    return jsonify({
        "status": "error",
        "code": 404,
        "message": "The requested resource was not found.",
    }), 404

@app.errorhandler(405)
def method_not_allowed(error):
    return jsonify({
        "status": "error",
        "code": 405,
```

```
        "message": "HTTP method not allowed for this endpoint.",
    }), 405

@app.errorhandler(500)
def internal_error(error):
    return jsonify({
        "status": "error",
        "code": 500,
        "message": "An internal server error occurred.",
    }), 500

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000, debug=True)
```

```
206   # 5 Error Handlers
207   @app.errorhandler(404)
208   def not_found(error):
209       return jsonify({
210           "status": "error",
211           "code": 404,
212           "message": "The requested resource was not found.",
213       }), 404
214
215   @app.errorhandler(405)
216   def method_not_allowed(error):
217       return jsonify({
218           "status": "error",
219           "code": 405,
220           "message": "HTTP method not allowed for this endpoint.",
221       }), 405
222
223   @app.errorhandler(500)
224   def internal_error(error):
225       return jsonify({
226           "status": "error",
227           "code": 500,
228           "message": "An internal server error occurred.",
229       }), 500
230
231   if __name__ == "__main__":
232       app.run(host="0.0.0.0", port=5000, debug=True)
```

Step 6 - Test with curl

Open a second PowerShell terminal while `python sensor_api/app.py` is running in the first.

```
python sensor_api/app.py
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> python sensor_api/app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.66.181.201:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 256-995-261
```

6a Health Check

Verify the API is online and returning metadata.

```
curl.exe -s http://localhost:5000/api/v1/health | python -m json.tool
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s http://localhost:5000/api/v1/health | python -m json.tool
{
    "service": "sensor-data-api",
    "status": "ok",
    "timestamp": "2026-05-15T02:51:44.813485+00:00",
    "version": "1.0"
}
```

6b List Sensors

Retrieve the list of active sensor types.

```
curl.exe -s http://localhost:5000/api/v1/sensors | python -m json.tool
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s http://localhost:5000/api/v1/sensors | python -m json.tool
{
    "count": 3,
    "data": [
        "pressure",
        "temperature",
        "humidity"
    ],
    "status": "success"
}
```

A JSON response containing a `"data"` array of sensor types (e.g., `["temperature", "humidity"]`). Successfully queries Parquet files via `lake_utils.py`.

6c Latest Reading

Retrieve real-time reading from Kafka, and test 404 logic.

```
curl.exe -s http://localhost:5000/api/v1/sensors/temperature/latest | python -m json.tool
curl.exe -s http://localhost:5000/api/v1/sensors/radar/latest | python -m json.tool
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s http://localhost:5000/api/v1/sensors/temperature/latest | python -m json.tool
{
    "data": {
        "sensor": "temperature",
        "timestamp": 1778812858712,
        "unit": "C",
        "value": 32.55
    },
    "status": "success"
}
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s http://localhost:5000/api/v1/sensors/radar/latest | python -m json.tool
{
    "message": "Unknown sensor type 'radar'. Valid types: ['humidity', 'pressure', 'temperature']",
    "status": "error"
}
```

```
127.0.0.1 - - [15/May/2026 11:12:09] "GET /api/v1/sensors/temperature/latest HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2026 11:12:14] "GET /api/v1/sensors/radar/latest HTTP/1.1" 404 -
```

The first returns a JSON object with reading data (Status 200). The second returns `{"status": "error", "message": "Unknown sensor type 'radasr'..."}` (Status 404).

Path validation is successfully rejecting unauthorized sensor types.

6d Statistics with Query Parameter

Test the query parameter filtering and validation logic.

```
curl.exe -s "http://localhost:5000/api/v1/sensors/temperature/stats?days=3" | python -m json.tool
curl.exe -s "http://localhost:5000/api/v1/sensors/temperature/stats?days=abc" | python -m json.tool
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s "http://localhost:5000/api/v1/sensors/temperature/stats?days=3" | python -m json.tool
{
    "count": 3,
    "data": [
        {
            "day": 15,
            "event_time": "Fri, 15 May 2026 10:40:48 GMT",
            "is_anomaly": null,
            "month": 5,
            "offset": 1,
            "partition": 0,
            "sensor_type": "temperature",
            "unit": "C",
            "value": 31.16,
            "year": 2026
        },
        {
            "day": 15,
            "event_time": "Fri, 15 May 2026 10:40:48 GMT",
            "is_anomaly": null,
            "month": 5,
            "offset": 2,
            "partition": 0,
            "sensor_type": "temperature",
            "unit": "C",
            "value": 34.78,
            "year": 2026
        },
        {
            "day": 15,
            "event_time": "Fri, 15 May 2026 10:40:49 GMT",
            "is_anomaly": null,
            "month": 5,
            "offset": 3,
            "partition": 0,
            "sensor_type": "temperature",
            "unit": "C",
            "value": 19.14,
            "year": 2026
        }
    ],
    "days": 3,
    "sensor_type": "temperature",
    "status": "success"
}
```

```
(bigdata) PS D:\EFREI\Data_Engineering\LAB\Lab5> curl.exe -s "http://localhost:5000/api/v1/sensors/temperature/stats?days=abc" | python -m json.tool
{
    "message": "Invalid 'days' parameter: invalid literal for int() with base 10: 'abc'",
    "status": "error"
}
```

```
127.0.0.1 - - [15/May/2026 11:13:28] "GET /api/v1/sensors/temperature/stats?days=3 HTTP/1.1" 200 -
127.0.0.1 - - [15/May/2026 11:13:59] "GET /api/v1/sensors/temperature/stats?days=abc HTTP/1.1" 400 -
```

First command returns aggregated data for 3 days (Status 200). Second command returns a 400 Bad request stating `Invalid`.

The API correctly casts to integers and catches `ValueError`.

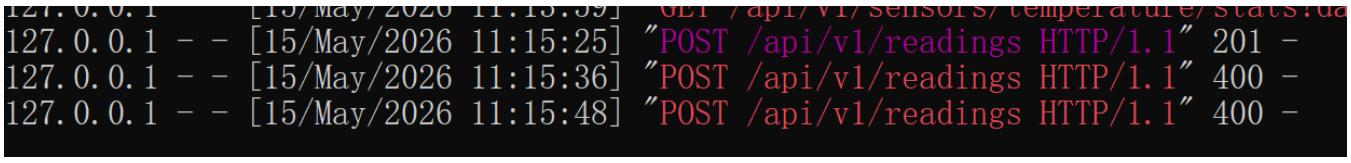
6e POST a New Reading

Verify that the API accepts JSON payloads, publishes to Kafka, and properly handles validation rules.

```
# Successful POST
curl.exe -s -X POST http://localhost:5000/api/v1/readings -H "Content-Type:
application/json" -d '{\"sensor\": \"temperature\", \"value\": 29.3, \"unit\": \"C\"}' |
python -m json.tool

# Missing field (400)
curl.exe -s -X POST http://localhost:5000/api/v1/readings -H "Content-Type:
application/json" -d '{\"sensor\": \"temperature\"}' | python -m json.tool

# Wrong Content-Type (400)
curl.exe -s -X POST http://localhost:5000/api/v1/readings -d "sensor=temperature&value=29.3"
| python -m json.tool
```



Success yields `201 Created` with partition/offset metadata. Errors return appropriate 400 JSON payloads.

Write pipeline is secured and functioning as intended.

Reflection Questions

1. **A client sends `GET /api/v1/sensors/temperature/stats` with no days parameter. The default is 7. Is it correct to return 200 or 400? Justify your answer.**

   It is correct to return a `200 OK`. The `days` parameter is a query parameter, which is meant to be optional for filtering collections. Since the API is designed to assign a sensible default value (7 days) when the parameter is absent, the request is valid and understood by the server, satisfying the criteria for a 200 success response.

2. **Explain why `POST /readings` is not idempotent but `PUT /readings/42` (replacing reading 42) is. Give a concrete scenario where this distinction matters in a retry mechanism.**

`POST /readings` is not idempotent because it is used to *create* a new resource; executing it multiple times will create multiple, duplicate resources. `PUT /readings/42` is idempotent because it replaces the specific resource entirely at that specific URI; calling it once or 100 times will result in the exact same server state. *Scenario:* If a network timeout occurs right after a client sends a request but before the response arrives, the client might trigger a retry. Retrying a `POST` could result in duplicate sensor readings in the database. Retrying a `PUT` is safe because it will just overwrite the resource with the exact same data again without creating duplicates.

3. **A colleague suggests returning 200 OK for all responses and putting the status code in the JSON body (e.g., `{"status": 404, ...}`). What are the problems with this approach?**

   This breaks standard HTTP and REST principles. Problems include:

   Caching mechanisms: Intermediate caches and proxies look at HTTP headers to determine cacheability. A 200 response tells the proxy to cache a request that was actually an error.

   Load Balancing & Routing: Load balancers rely on standard HTTP codes (like returning non-2xx statuses on a `/health` check) to remove failing instances from a pool.

   Client tool compatibility: Standard HTTP clients (browsers, libraries, `curl`) evaluate request success based on HTTP headers, not the JSON payload. The client would have to write custom logic to parse the body to discover if an operation actually succeeded.

4. **What is the difference between a 400 Bad Request and a 422 Unprocessable Entity? Give one example of each for the `POST /readings` endpoint.**

   400 Bad Request indicates a request with bad or malformed syntax. Example: Sending invalid JSON or a payload missing the required `sensor` field.

   422 Unprocessable Entity is used when the payload is syntactically valid JSON, but it fails semantic business validation. Example: Passing valid JSON but assigning a string to `value` instead of a float, or providing an invalid sensor type like `"radar"`.

5. **The `GET /sensors/temperature/latest` endpoint queries Kafka. If the Kafka cluster is down, what HTTP status code should the API return? What should the response body look like?**

   The API should return a `503 Service Unavailable` (or `500 Internal Server Error`) indicating the server is temporarily failing to fulfill the request due to backend unreachability. The response body must *not* contain internal error details like stack traces, to prevent security vulnerabilities. It should be a structured, generic JSON response:

```
{
    "status": "error",
    "code": 503,
    "message": "Service temporarily unavailable. Failed to connect to data source."
}
```